Environmental safety monitor
Wiring and Proteus guide

Working hardware revision | 26 October 2026 | AT89S52, 11.0592 MHz

Owner reports successful LCD, ADC, gas/buzzer and temperature/fan operation. This guide preserves the assigned application pins and current HEX. Keypad removal has faculty approval. No Proteus project has been created or run for this guide.

What replaces the development board?

Use a standalone AT89S52 symbol with 5 V supply, crystal and reset circuitry. The NSK board already provides these functions in hardware; do not add a second crystal or reset circuit to the real board. RS232/MAX232 and the ISP programmer are not needed to execute a HEX in Proteus.

Block	Actual hardware	Proteus representation / search term
Controller	NSK board + AT89S52	AT89S52 VSM model; verify installed support
ADC and clock	ADC0808 + NE555	ADC0808 / ADC0808-N; NE555
LCD	1602 with HW-61 backpack	PCF8574 + compatible 16x2 HD44780 LCD (e.g. LM016L)
Temperature	LM35	LM35 model; or controlled voltage for interface testing
Gas	MQ-2 analogue signal	0-5 V adjustable source/potentiometer labelled MQ-2 AO surrogate
Outputs	Two-channel LOW-trigger relay module	Two active-LOW driver/relay equivalents; see page 6
Loads	5 V motor and active buzzer	DC motor; active DC buzzer or labelled indicator

Supply rules

On hardware, the 12 V adapter enters the NSK board barrel socket. Only regulated 5 V goes to the MCU, ADC, NE555, LCD and 5 V modules. Supply heater, relays and motor from a suitably rated 5 V source; its rating must cover starting current. Join grounds, but do not parallel independent regulated 5 V outputs.

For an initial functional Proteus schematic, use an ideal +5 V rail and GND. Label motor power separately if using a separate supply. This simplification does not test real regulator heating, supply sag or motor interference.

Scope: exact application wiring is recorded here. Board internals, relay-module internals and unseen component values are not claimed to have been reverse-engineered. Suggested simulation circuits are explicitly identified.

AT89S52: complete 40-pin map

Pin	Port/function	Connection
1	P1.0/T2	Unused by firmware; NSK user LED attached
2	P1.1/T2EX	Unused
3	P1.2	LCD SDA
4	P1.3	LCD SCL
5	P1.4	Unused
6	P1.5/MOSI	ISP only; disconnect programmer to run
7	P1.6/MISO	ISP only
8	P1.7/SCK	ISP only
9	RST	Active-HIGH reset; standalone circuit below
10	P3.0/RXD	Unused by firmware; NSK RS232 circuitry
11	P3.1/TXD	Unused by firmware; NSK RS232 circuitry
12	P3.2	ADC Address A, ADC pin 25
13	P3.3	ADC START pin 6 + ALE pin 22
14	P3.4	Relay module IN1: fan/motor, LOW=ON
15	P3.5	Relay module IN2: buzzer, LOW=ON
16	P3.6	ADC EOC, ADC pin 7
17	P3.7	Unused
18	XTAL2	11.0592 MHz crystal to pin 19
19	XTAL1	11.0592 MHz crystal to pin 18
20	GND	Common GND
21	P2.0	ADC D0, ADC pin 17
22	P2.1	ADC D1, ADC pin 14
23	P2.2	ADC D2, ADC pin 15
24	P2.3	ADC D3, ADC pin 8
25	P2.4	ADC D4, ADC pin 18
26	P2.5	ADC D5, ADC pin 19
27	P2.6	ADC D6, ADC pin 20
28	P2.7	ADC D7, ADC pin 21
29	PSEN	Leave unconnected; internal program memory
30	ALE/PROG	Leave unconnected; NOT ADC ALE
31	EA/VPP	+5 V for internal Flash execution
32	P0.7	Unused; keypad removed
33	P0.6	Unused; keypad removed
34	P0.5	Unused; keypad removed
35	P0.4	Unused; keypad removed
36	P0.3	Unused; keypad removed
37	P0.2	Unused; keypad removed
38	P0.1	Unused; keypad removed
39	P0.0	Unused; keypad removed
40	VCC	Regulated +5 V

Standalone Proteus support: crystal across 18-19; 30 pF from each crystal pin to GND (starting values). 10 kOhm RST-to-GND, 10 uF capacitor from +5 V to RST (+ at +5 V), reset button from RST to +5 V. 100 nF across 40-20. Set model clock property to 11.0592 MHz. These are proposed standalone values, not measured NSK values.

Pin audit: 15 application GPIO pins, each assigned once. The intentional shared START/ALE net uses one output to drive two ADC inputs. ISP pins do not overlap application signals. No Port 0 pull-ups are needed for unused standalone pins.

ADC0808: complete 28-pin map

Pin	Function	Connection
1	IN3	GND
2	IN4	GND
3	IN5	GND
4	IN6	GND
5	IN7	GND
6	START	MCU P3.3 / pin 13; also ADC pin 22
7	EOC	MCU P3.6 / pin 16
8	D3	MCU P2.3 / pin 24
9	OE	+5 V
10	CLOCK	NE555 output pin 3
11	VCC	+5 V
12	VREF+	+5 V
13	GND	Common GND
14	D1	MCU P2.1 / pin 22
15	D2	MCU P2.2 / pin 23
16	VREF-	GND
17	D0, LSB	MCU P2.0 / pin 21
18	D4	MCU P2.4 / pin 25
19	D5	MCU P2.5 / pin 26
20	D6	MCU P2.6 / pin 27
21	D7, MSB	MCU P2.7 / pin 28
22	ALE	MCU P3.3 / pin 13; also ADC pin 6
23	Address C	GND
24	Address B	GND
25	Address A	MCU P3.2 / pin 12
26	IN0	LM35 output
27	IN1	MQ-2 module AO
28	IN2	GND

Reference, data and handshake

Fit 100 nF between pins 11-13 and 100 nF between 12-16. Inputs IN0/IN1 remain connected to their sensors; never ground an input while its sensor output is connected. Unused analogue inputs are grounded.

Do not wire the eight data pins in physical numerical order. Some symbols label output significance as 2^-1 through 2^-8: 2^-1 is D7/MSB, and 2^-8 is D0/LSB. Follow significance and package pin numbers.

Firmware sets the address, pulses the shared START/ALE net, checks EOC LOW then HIGH with timeouts, and reads Port 2. OE is permanently HIGH. ADC ALE is driven by P3.3, not by the MCU ALE pin 30.

NE555 clock and analogue sensors

NE555 pin	Function	Connection
1	GND	GND
2	TRIG	Join pin 6; timing junction
3	OUT	ADC pin 10 CLOCK
4	RESET	+5 V
5	CONTROL	10 nF capacitor (103) to GND
6	THRESH	Join pin 2
7	DISCH	Junction of RA and RB
8	VCC	+5 V

Component	Value	Endpoints
RA	10 kOhm	+5 V to NE555 pin 7
RB	10 kOhm	NE555 pin 7 to joined pins 2/6
Timing C	1 nF (102)	Pins 2/6 to GND
Control C	10 nF (103)	Pin 5 to GND
Decoupling	100 nF (104)	Pins 8 to 1
Bulk capacitor	10 uF	+ to +5 V; - to GND

Nominal clock = 1.44 / [(RA + 2RB) x C] = 48 kHz; duty approximately 67%. Put a virtual oscilloscope at ADC pin 10 and verify frequency and HIGH/LOW voltage.

Known limitation: at 5 V the ADC HIGH requirement is 3.5 V. A bipolar NE555 does not guarantee this HIGH level. The owner reports operation, but that does not establish noise margin. Preserve the current wiring in the schematic and check the model; do not silently replace the clock or increase its supply. A buffer/CMOS 555 would be a separately documented change if required.

Sensor connections

Device	Supply / ground	Signal
LM35, confirmed TO-92 only	Pin 1 to +5 V; pin 3 to GND	Pin 2 output to ADC pin 26
MQ-2 breakout module	Labelled VCC to +5 V; GND common	AO to ADC pin 27; DO unused

Verify LM35 package orientation against its manufacturer drawing; package pin numbering is not a breadboard left/right assumption. A bare six-pin MQ-2 needs a separate heater/load circuit; module labels above do not apply to it.

In Proteus, use the LM35 model if available. Use an adjustable 0-5 V source or 10 kOhm pot (ends to 5 V/GND, wiper to ADC27) as the gas AO surrogate. Label it clearly: this tests response to voltage, not gas chemistry or ppm calibration. At 5 V reference one count is 19.53 mV, approximately 1.95 C with LM35.

LCD: reproduce the I2C backpack

Keep the existing HEX: a parallel LCD wired straight to the MCU will not work with it. Reproduce the working HW-61 mapping with a PCF8574 plus a compatible 16x2 LCD. The hardware needs only its existing four backpack wires.

Backpack terminal	MCU / supply
GND	Common GND
VCC	+5 V
SDA	P1.2 / MCU pin 3
SCL	P1.3 / MCU pin 4

PCF8574 pin	Function	Proteus connection
1	A0	+5 V
2	A1	+5 V
3	A2	+5 V
4	P0	LCD RS pin 4
5	P1	LCD RW pin 5
6	P2	LCD E pin 6
7	P3	Backlight control; reserve, not LCD data
8	VSS	GND
9	P4	LCD D4 pin 11
10	P5	LCD D5 pin 12
11	P6	LCD D6 pin 13
12	P7	LCD D7 pin 14
13	INT	Unconnected
14	SCL	MCU P1.3 / pin 4
15	SDA	MCU P1.2 / pin 3
16	VDD	+5 V

Add 4.7 kOhm from SDA to +5 V and 4.7 kOhm from SCL to +5 V in the standalone simulation. Add 100 nF across PCF pins16-8. A0/A1/A2 HIGH gives 7-bit address0x27 for PCF8574, or0x3F for PCF8574A. This is a simulation choice; actual strap state was not measured. Current firmware scans both address ranges.

LCD pin(s)	Connection
1 VSS / 2 VDD	GND / +5 V
3 VO/VEE	Wiper of 10 kOhm contrast pot; outer ends +5 V/GND
4 RS / 5 RW / 6 E	PCF8574 pins 4 / 5 / 6 respectively
7-10 D0-D3	Unused in 4-bit mode; leave open
11-14 D4-D7	PCF8574 pins 9 / 10 / 11 / 12
15 A / 16 K, if present	Backlight only; see note below

Many LCD simulation symbols omit backlight pins. If present, a fixed backlight may be represented by +5 V through220 Ohm to A, K toGND; leave PCF P3 unused. Label this as an always-on simulation simplification. Do not drive a physical backlight directly from PCF P3.

Relays, motor and buzzer

Actual module connections - no new hardware drivers

Connection	Endpoint
Module VCC / GND	Regulated +5 V / common GND
IN1	P3.4 / MCU pin14; LOW=K1 ON
IN2	P3.5 / MCU pin15; LOW=K2 ON
K1 COM	Motor +5 V supply
K1 NO	Motor positive terminal
Motor negative	Motor supply GND
K1 NC	Unconnected
K2 COM	Regulated +5 V
K2 NO	Active buzzer positive
Buzzer negative	GND
K2 NC	Unconnected

Use the module terminal labels, not an assumed terminal order. Retain its existing jumper position. The physical module contains its own coil drivers; never connect a bare coil or the two-pin buzzer directly to an MCU output.

Recommended motor suppression: 1N4007 across motor, stripe/cathode at motor+, anode at motor-. Add100 nF across motor terminals and a starting470 uF bulk capacitor across its supply, observing polarity. These were recommended; their actual installation has not been independently confirmed.

Proteus substitute if the relay module model is unavailable

Represent each channel with a PNP transistor and a 5 V SPDT relay. This is a proposed functional equivalent, NOT the measured module internal schematic. It preserves LOW=ON without changing firmware. Validate the selected transistor/relay models before calling the simulation complete.

Per channel part / terminal	Proposed simulation connection
PNP BC327 emitter	+5 V
PNP base	MCU output through1 kOhm; also10 kOhm base-to-emitter
PNP collector	Relay coil upper terminal
Relay coil lower terminal	GND
Coil model	5 V rated; use a compatible coil, e.g.100 Ohm (50mA)
Coil flyback diode	1N4007 cathode to collector, anode toGND
Contact COM/NO/NC	As actual contact table above

Repeat for K1/P3.4 and K2/P3.5. Use transistor E/B/C symbol labels, not an assumed package pinout. Use an active DC buzzer model; if unavailable, a labelled lamp/logic indicator can verify its switched supply but does not prove sound output.

Build order and simulation checks

Not started: the following is a proposed workflow. Installed VSM device models and licence availability must be checked before promising a runnable project.

1. Place the AT89S52, power/reset/crystal support from page2, then set its clock property to11.0592MHz. Connect EA HIGH. Verify hidden power pins resolve to the correct +5V/GND nets.

2. Place the PCF8574 and LCD from page5. Load the existing final HEX into the MCU Program File property. No USB programmer or RS232 transfer is involved in simulation.

3. Add ADC0808 and wire every pin from page3. Use labelled nets for D0-D7, ADC_A, START_ALE and EOC. Check pin significance rather than visual symbol ordering.

4. Add the NE555 oscillator from page4. Observe ADC CLOCK with the virtual oscilloscope. If clock levels fail, record that result rather than silently substituting an ideal clock.

5. Add LM35 and gas AO voltage surrogate. Start with25C and gas AO0.5V. Add output-equivalent relay circuits and the loads from page6.

6. Run normal, temperature, gas and fault tests below. Hold inputs long enough for filtering and three qualifying batches. Save the schematic and evidence only after checking actual results.

Test stimulus	Expected firmware response
LM35 25C; gas AO0.5V	About25C, gas raw26; both outputs OFF after settling
Temperature raised to45C	Fan ON after confirmation; buzzer follows gas only
Temperature lowered to30C	Fan OFF after confirmation
Gas AO raised to3.0V	Raw about154; buzzer ON after confirmation
Gas AO lowered to1.5V	Raw about77; buzzer OFF after confirmation
Stop ADC clock	Conversion timeout; motor OFF, buzzer ON; fault latched
Correct fault then reset	Restart; both outputs initially OFF

Current firmware: five-sample median; three qualifying batches; nominal fan40C ON/36C OFF (actual raw21/18); gas raw140 ON/100 OFF. No gas warmup inhibition. Timer0/Timer1 polling; external NE555 clock. Do not describe this version as interrupt-driven.

LCD first line: **Temp:025C G:090**. Second line: **R:013 F:0 B:0**. R is filtered temperature ADC raw; F/B are output commands, not contact feedback. Temperature is already converted to Celsius.

Limits, programming and viva notes

Working HEX and physical ISP

Use the latest file under the project folder:
firmware/environment_monitor/environment_monitor/environment_monitor.hex
Build verified without compiler warnings/errors. The owner reports successful integrated operation after the LCD text update. Quantitative accuracy and supply/clock margins were not independently measured.

Programmer printed pin	NSK ISP / MCU connection
1 MOSI	MOSI / P1.5 pin6
9 MISO	MISO / P1.6 pin7
7 SCK	SCK / P1.7 pin8
5 RST	RST / pin9
10 GND	GND / pin20
2 VCC	Avoid joining independent power outputs; retain verified working power arrangement
3 NC; other GND pins	NC open; additional grounds optional

ProgISP1.72 recognises the programmer as ZF-008 HV Programmer. The supplied cable caused errors; individual jumper wires worked. AT89S52 signature1E5206. Pin numbers follow the programmer label, not an assumed connector viewing direction. For Proteus, none of this programmer hardware is required.

What simulation cannot establish

Gas concentration/calibration, heater conditioning, actual airflow, contact arcing, reliable immunity to motor noise, real regulator capacity and sensor accuracy require physical evidence. Earlier temperature jumps and overlapping gas readings were reported; later functional success does not establish calibration. Do not use lighter gas near switching contacts. The direct NE555 clock margin remains a documented limitation.

Five viva questions

Why an external ADC? AT89S52 has no built-in analogue-to-digital converter. ADC0808 provides8-bit measurements.

Why is NE555 needed? It provides the ADC conversion clock; the MCU crystal separately clocks program execution.

How is Celsius calculated? With5V reference, temperature is raw x500/256; resolution is about1.95C/count.

Why separate ON and OFF thresholds? Hysteresis and repeated readings reduce relay chatter near a threshold.

Why PCF8574 in Proteus? It recreates the I2C backpack so the existing two-wire LCD firmware can run unchanged.

Primary references

AT89S52 datasheet

ADC0808 datasheet

NE555 datasheet

LM35 datasheet

PCF8574 datasheet